BestRun EV

Apex Pro Battery & Drive Technology Guide

Technical Reference for Authorized Dealers - Apex Series Launch

Introduction

The Apex Pro introduces BestRun EV's first production use of Lithium Iron Phosphate (LFP) battery chemistry and the proprietary BRD-750 mid-drive motor system. This guide provides dealers with the technical background needed to confidently present and service the Apex Pro.

1. LFP Battery Technology

1.1 Why LFP?

Previous BestRun EV models used NMC (Nickel Manganese Cobalt) chemistry. The Apex Pro switches to LFP for three reasons:

- Thermal stability: LFP cells do not enter thermal runaway under the same conditions as NMC, making them safer for high-performance use and storage in warm climates.
- Cycle life: LFP delivers 2,000+ full charge cycles before reaching 80% capacity retention, versus 800-1,000 for typical NMC. At one charge per day, this is over 5 years of daily use.
- Flat discharge curve: LFP maintains a near-constant voltage (~3.2 V/cell) through 80% of its discharge, delivering more consistent power output than NMC.

1.2 Battery Pack Specifications

Parameter	Value
Chemistry	LFP (LiFePO4)
Nominal Voltage	48 V
Capacity	25 Ah
Energy	1,200 Wh
Cell Configuration	16S2P (16 series, 2 parallel)
Cell Supplier	CATL (tier 1 EV supplier)
Nominal Cell Voltage	3.2 V
Max Charge Voltage	3.65 V/cell (58.4 V pack)
Min Discharge Voltage	2.8 V/cell (44.8 V pack)
Max Continuous Discharge Current	60 A
Peak Discharge Current (5 s)	120 A
Operating Temperature	-20 deg C to +60 deg C (discharge); 0 to +45 deg C (charge)
Storage Temperature	-20 deg C to +35 deg C recommended
IP Rating	IP67
Weight	4.5 kg
Dimensions	420 x 95 x 75 mm (integrated downtube)

1.3 Charging System

The Apex Pro ships with a dedicated 8A fast charger (BestRun EV Part No. CHR-8A-48V). The Battery Management System (BMS) handles all charge termination.

Charger	Input	Output	0-80%	0-100%
Fast Charger (included)	100-240V AC, 50/60Hz	54.4V / 8A (435W)	~2.5 h	~3.5 h
Standard Charger (optional)	100-240V AC, 50/60Hz	54.4V / 4A (218W)	~5.0 h	~6.5 h
Dual-port Fast Charger (opt.)	100-240V AC, 50/60Hz	54.4V / 12A (650W)	~1.8 h	~2.5 h

Note: Charging below 0 deg C is not supported. The BMS will refuse to accept charge current if the battery temperature is below 0 deg

C. Allow the battery to warm up indoors before charging. Charging above 45 deg C will throttle input current to protect cell longevity.

1.4 Range Calculations

The 200 km Eco Mode range is calculated at 6 Wh/km (WLTP Urban, flat terrain, 75 kg rider, 20 deg C). The table below shows estimated range under common real-world conditions:

Condition	Wh/km	Eco Range	Standard Range
WLTP Urban (reference)	6.0	200 km	140 km
Mixed urban/suburban	7.5	160 km	112 km
Hilly terrain (+200m/100km)	9.0	133 km	93 km
Heavy off-road (trail)	12.0	100 km	70 km
Cold weather (0 deg C)	8.5	141 km	99 km
Heavy rider (100 kg)	7.0	171 km	120 km

2. BRD-750 Mid-Drive Motor

2.1 Motor Overview

The BRD-750 is BestRun EV's first proprietary mid-drive motor, developed in partnership with the BestRun EV R&D centre in Munich. Mid-drive placement (at the bottom bracket) provides superior weight distribution and allows the rider to use the bike's own gearing for optimal efficiency across all terrain.

Parameter	Value
Rated Power	750 W (EU pedelec); 1,000 W (non-EU markets)
Peak Power (5 s)	1,200 W
Rated Torque	95 Nm
Peak Torque	120 Nm
Cadence Sensor	12-magnet cadence + torque sensor (dual)
Gear Sensor	Integrated automatic gear shift detection
Efficiency (peak)	92%
Weight	3.2 kg
Bottom Bracket Standard	73 mm threaded (BSA)
Noise Level	<55 dB(A) at rated load

2.2 Assist Modes

Mode	Max Motor Contribution	Typical Use Case
Eco	35% of rider input	Long-distance commuting, maximum range
Standard	100% of rider input	Everyday mixed riding
Sport	200% of rider input	Hills, headwinds, moderate off-road
Turbo	340% of rider input	Steep climbs, full off-road performance
Walk Assist	6 km/h constant	Pushing the bike up ramps or steep sections

3. Battery Management System (BMS)

The integrated BMS monitors and protects the battery at all times. Key functions:

- Cell balancing: active balancing during charge and passive balancing during storage
- Overtemperature protection: cuts charge/discharge above 60 deg C cell temperature
- Overcurrent protection: limits peak current to 120 A; trips at 130 A
- Deep discharge protection: locks battery above 2.8 V/cell to prevent irreversible capacity loss
- State of health (SoH) reporting: available via BestRun EV Pro app (requires Bluetooth connection)
- Firmware updatable: BMS firmware updates delivered OTA via BestRun EV Pro app

4. Service Intervals & Dealer Obligations

Service Item	Interval	Dealer Action
Drive system inspection	Every 5,000 km or 12 months	Check motor mount torque, clean drive unit
Battery capacity check	Every 12 months	Run BestRun EV Diagnostic Tool, record SoH
BMS firmware update	When notified via app	Connect laptop, run BestRun EV Service Tool
Battery replacement	When SoH < 70%	Order via BestRun EV Parts Portal (BAT-PRO-LFP)
Motor replacement	On failure or >8,000 km if abnormal r	BRD-750 is dealer-replaceable; requires certified training
Charging port inspection	Every 5,000 km	Check for corrosion, ensure IP67 seal intact

Note: The BRD-750 motor and LFP battery require BestRun EV Apex-Certified Technician training before dealer-level service. Enrol via the dealer portal at bestrun-ev.com/dealer/training.